

Cloud-Based Data Processing

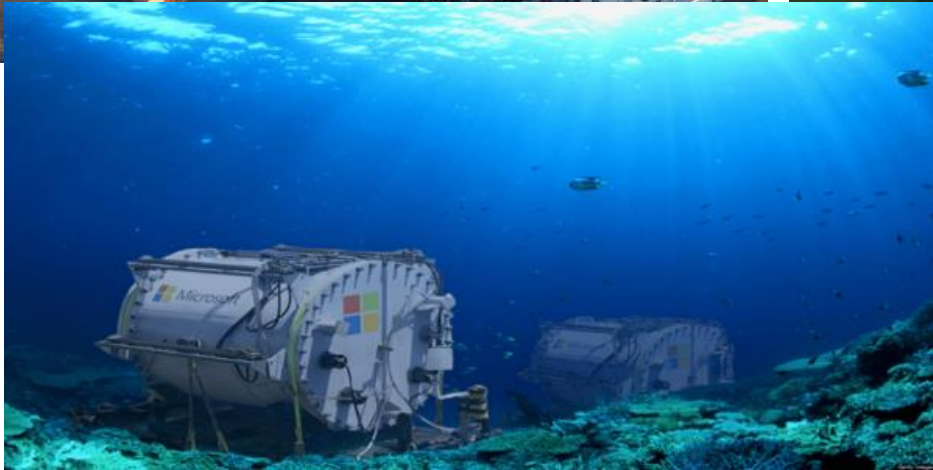
Data Centers

Data Centers

- Data center (DC) is a physical facility that enterprises use to house computing and storage infrastructure in a variety of networked formats.
- Main function is to deliver utilities needed by the equipment and personnel:
 - Power
 - Cooling
 - Shelter
 - Security
- Size of typical data centers:
 - 500 – 5000 sqm buildings
 - 1 MW to 10-20 MW power (avg 5 MW)



Example data centers



Datacenters around the globe



Modern DC for the Cloud architecture

■ Geography:

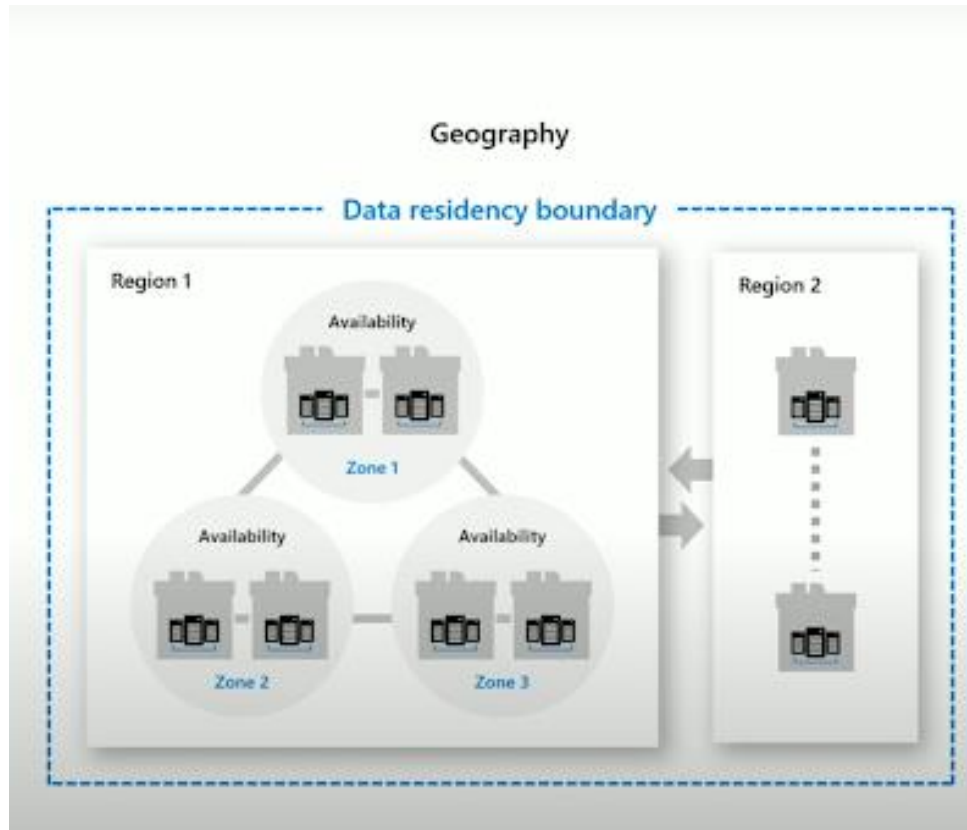
- Two or more regions
- Meets data residency requirements
- Fault-tolerant from complete region failures

■ Region:

- Set of datacenters within a metropolitan area
- Network latency perimeter < 2ms

■ Availability Zones:

- Unique physical locations within a region
- Each zone made up of one or more DCs
- Independent power, cooling, networking
- Inter-AZ network latency < 2ms
- Fault tolerance from DC failure



Data Centers

■ Traditional data centers

- Host a large number of relatively small- or medium-sized applications, each running on a dedicated hardware infrastructure that is decoupled and protected from other systems in the same facility
- Usually for multiple organizational units or companies

■ Modern data centers (a.k.a., Warehouse-scale computers)

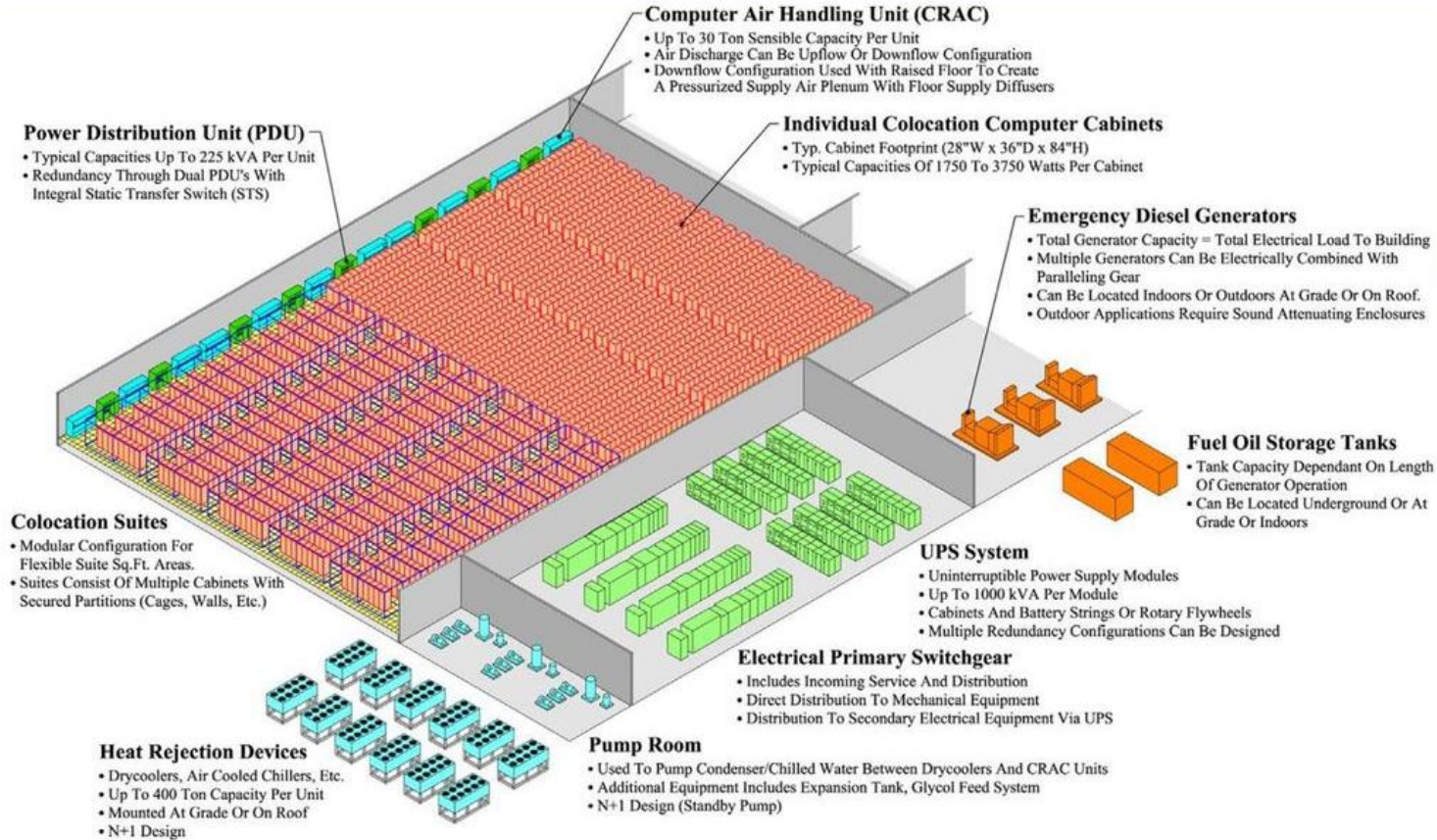
- Usually belong to a single company to run a small number of large-scale applications
 - Google, Facebook, Microsoft, Amazon, Alibaba, etc.
- Use a relatively homogeneous hardware and system software
- Share a common systems management layer
- Sizes can vary depending on needs

Datacenter Architecture

Scale-up vs. scale-out

- **Scale-up:** high cost powerful CPUs, more cores, more memory
 - **Scale-out:** adding more low cost, commodity servers
-
- Supercomputer vs. data center
 - **Scale**
 - Blue waters = 40K 8-core “servers”
 - Microsoft Chicago Data centers = 50 containers = 100K 8-core servers
 - **Network architecture** 
 - Supercomputers: InfiniBand, low-latency, high bandwidth protocols
 - Data Centers: (mostly) Ethernet based networks
 - **Storage**
 - Supercomputers: separate data farm
 - Data Centers: use disk on node + memory cache

Main components of a datacenter



Traditional Data Center Architecture

Servers mounted on 19" rack cabinets



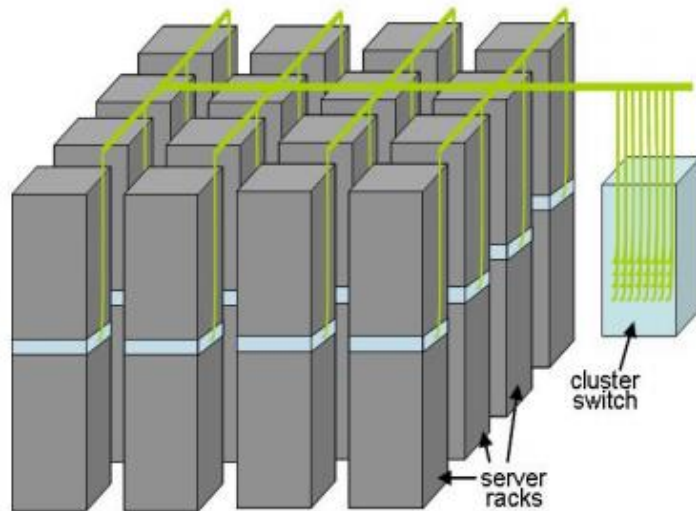
Servers

- CPUs
- DRAM
- Disks



Racks

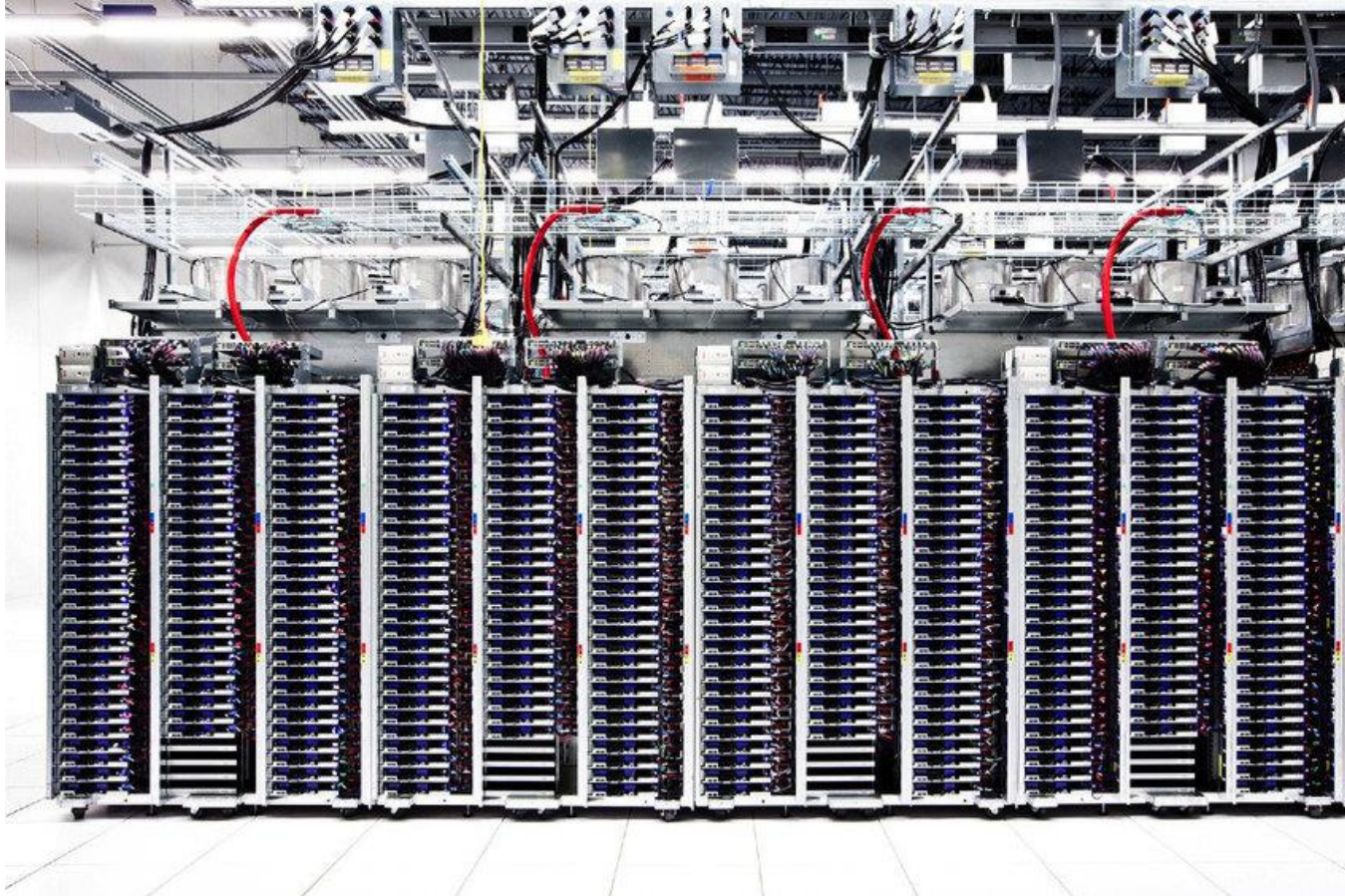
- 40-80 servers
- Ethernet switch



Clusters

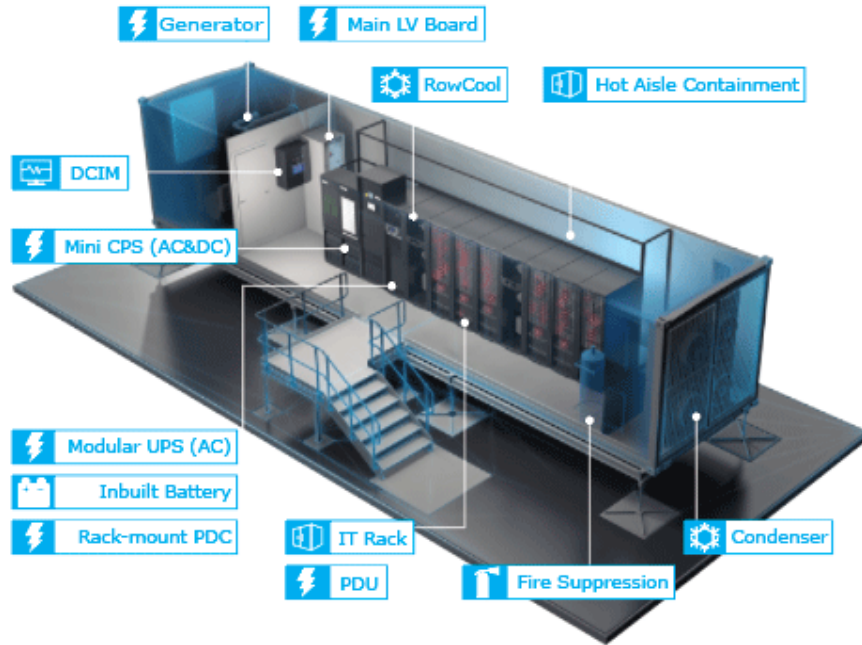
Racks are placed in single rows forming corridors between them.

A Row of Servers in a Google Data Center



- Src: the datacenter as a computer – an introduction to the design of warehouse-scale machines

Inside a modern data center



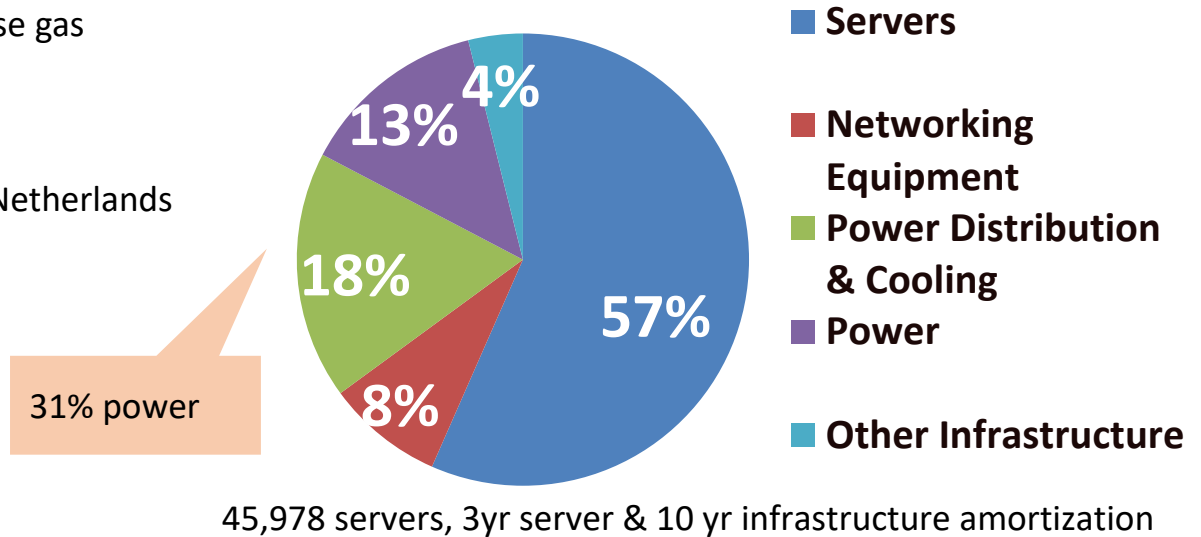
- Today's DC use shipping containers packed with 1000s servers each.
- For repairs, whole containers are replaced.



Costs for operating a data center

- DCs consume 3% of global electricity supply (416.2 TWh > UK's 300 TWh)
- DCs produce 2% of total greenhouse gas emissions
- DCs produce as much CO2 as The Netherlands or Argenti

Monthly cost = \$3'530'920



Power Usage Effectiveness (PUE)

- **PUE is the ratio of**

- The total amount of energy used by a DC facility
- To the energy delivered to the computing equipment

- **PUE is the inverse of data center infrastructure efficiency**

- **Total facility power** = covers **IT systems** (servers, network, storage) +
other equipment (cooling, UPS, switch gear, generators, lights, fans, etc.)

Achieving PUE

- **Location of the DC** – cooling and power load factor
- **Raise temperature of aisles**
 - Usually 18-20 C; Google at 27 C
 - Possibly up to 35 C (trade off failures vs. cooling costs)
- **Reduce conversion of energy**
 - E.g., Google motherboards work at 12V rather than 3.3/5V
- **Go to extreme environments**
 - Arctic circle (Facebook)
 - Floating boats (Google)
 - Underwater DC (Microsoft)
- **Reuse dissipated heat**

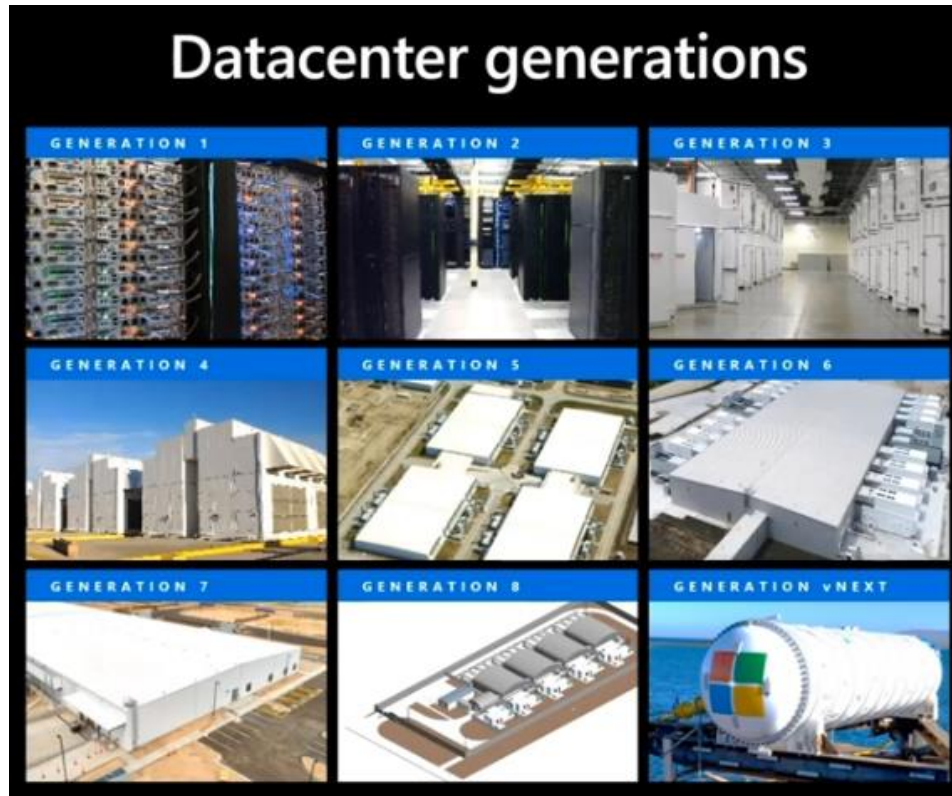
Price per Kilo Watt Hour	Where?	Possible Reason Why
3.6 cents	Idaho	Hydroelectric Power; Not Sent Long Distance
10.0 cents	California	Electricity Transmitted Long Distance over the Grid; Limited Transmission Lines in the Bay Area; No Coal Fired Electricity Allowed in California.
18.0 cents	Hawaii	Must Ship Fuel to Generate Electricity



Evolution of data center design

1989-2005	2007	2009	2012	2015
Generation 1	Generation 2	Generation 3	Generation 4	Generation 5
2.0+ PUE	1.4 – 1.6 PUE	1.2 – 1.5 PUE	1.12 – 1.20 PUE	1.07 – 1.19 PUE
 Colocation	 Density	 Containment	 Modular	 SW Defined
Server Capacity 20 year Technology	Rack Density & Deployment Minimized Resource Impact	Containers, PODs Scalability & Sustainability Air & Water Economization Differentiated SLAs	ITPACs & Colocations Reduced Carbon Right-Sized Faster Time-to-Market Outside Air Cooled	Fully Integrated Resilient Software Common Infrastructure Operational Simplicity Flexible & Scalable

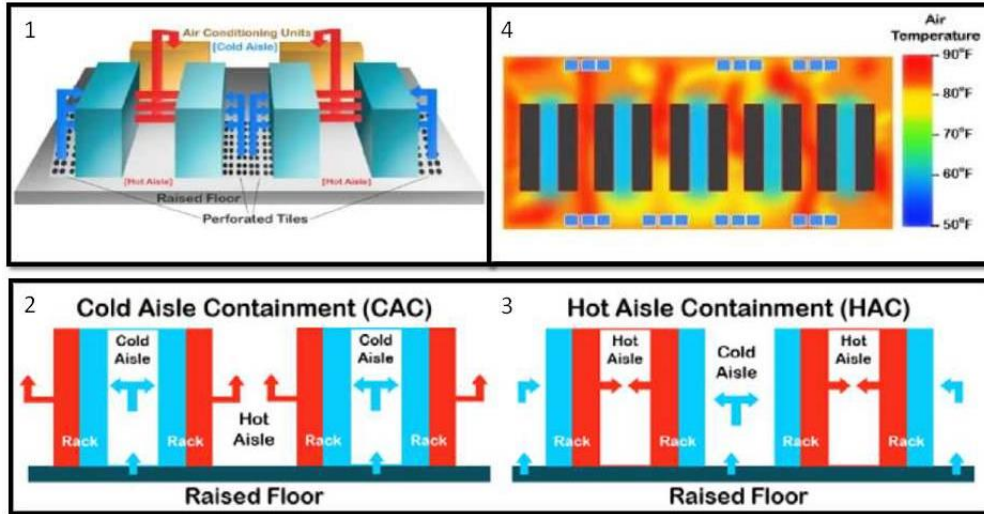
Evolution of datacenter design



- **Gen 6: scalable form factor (2017)**
 - Reduced infrastructure, scale to demand
 - 1.17-1.19 PUE
- **Gen 7: Ballard (2018)**
 - Design execution efficiency
 - Flex capacity enabled
 - 1.15-1.18 PUE
- **Gen 8: Rapid deploy datacenter (2020)**
 - Modular construction and delivery
 - Equipment skidding and preassembly
 - Faster speed to market
- **Project Natick (future) – 1.07 PUE or less**

Datacenter Challenges

Challenge 1: Cooling data centers



1- Conventional cooling
2- Cold Aisle Containment (CAC)

3- Hot Aisle Containment (HAC)
4- Thermal modelling

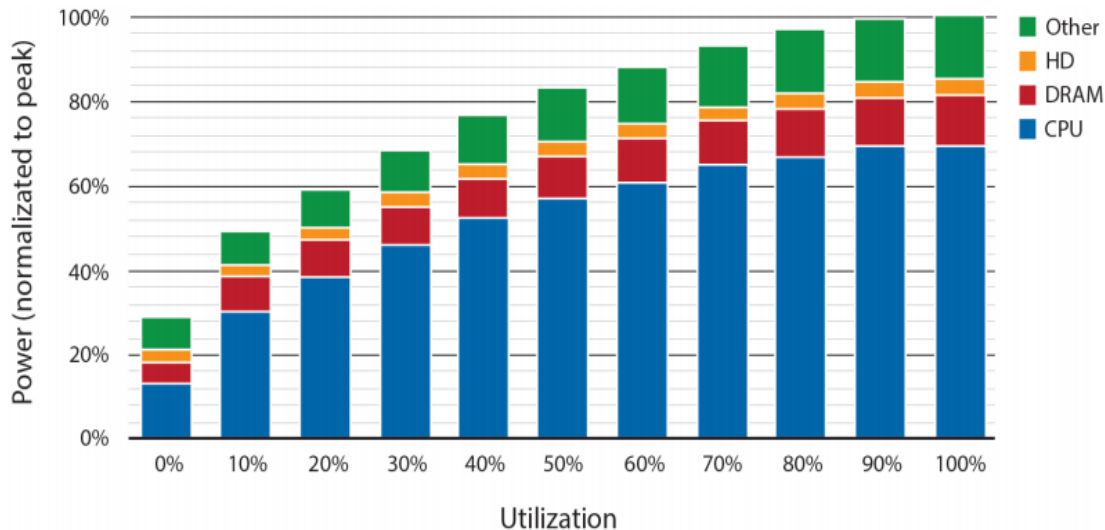
Cooling plant at a Google DC in Oregon



Challenge 2: Energy Proportional Computing

- **Average real world DC and servers are too inefficient.**
 - waste 2/3+ of their energy
- **Energy consumption is not proportional to the load**
 - CPUs are not so bad but the other components are
 - CPU is the dominant energy consumer in servers – using 2/3 of energy when active/idle.
- **Try to optimize workloads**
- **Virtualization and consolidation.**

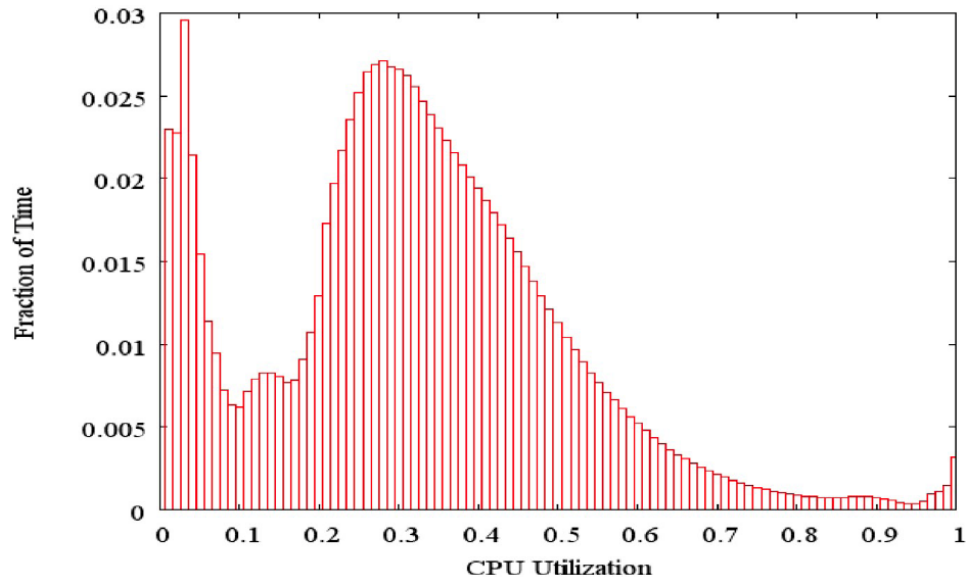
Sub-system power usage in an x86 server as the compute load varies from idle to full (reported in 2012).



• src: "The Datacenter as a Warehouse Computer"

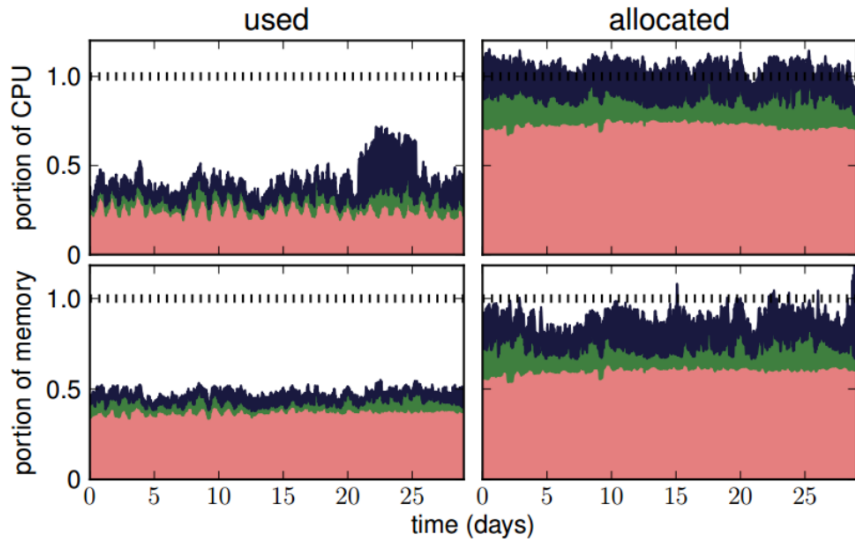
Challenge 3: Servers are idle most of the time

- For **non-virtualized servers** 6-15% utilization
- **Server virtualization** can **boost** to an average **30% utilization**
- **Need** for **resource pooling** and application and server **consolidation**
- Need for **resource virtualization**



src: Luiz Barroso, Urs Hölzle "The Datacenter as a Computer"

Challenge 4: Efficient monitoring



- Even with virtualization and software defined DC, **resource utilization can be poor.**
- Need for **efficient monitoring** (measurement) and **cluster management**.
- Goal to **meet SLOs and SLIs**.
- **Job's tail latency matters!**

Improving resource utilization

■ Hyper-scale system management software

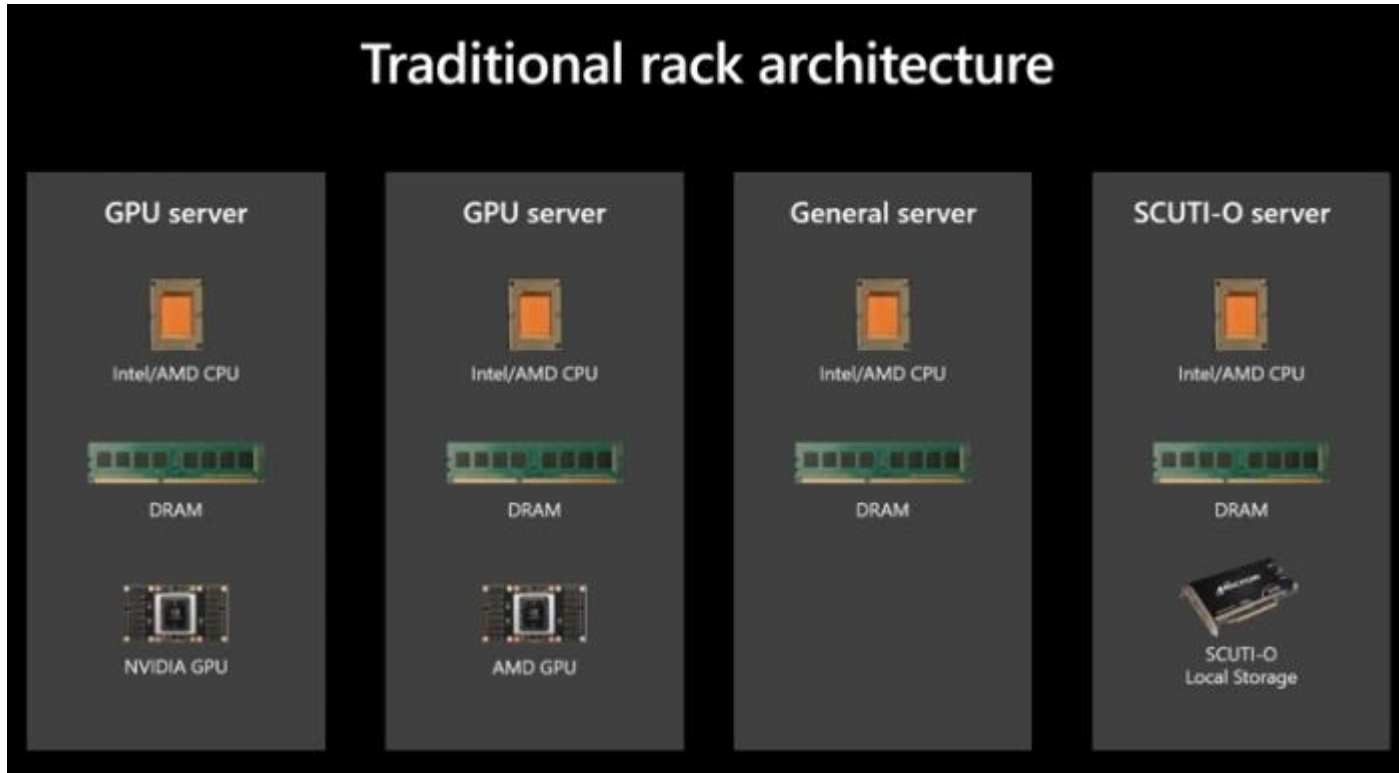
- Treat the datacenter as a warehouse scale computer
- Software defined datacenters
- System software that allows DC operations to manage the entire DC infrastructure
- Compose a system using pooled resources of compute, network, and storage based on workload requirement

■ Dynamic resource allocation

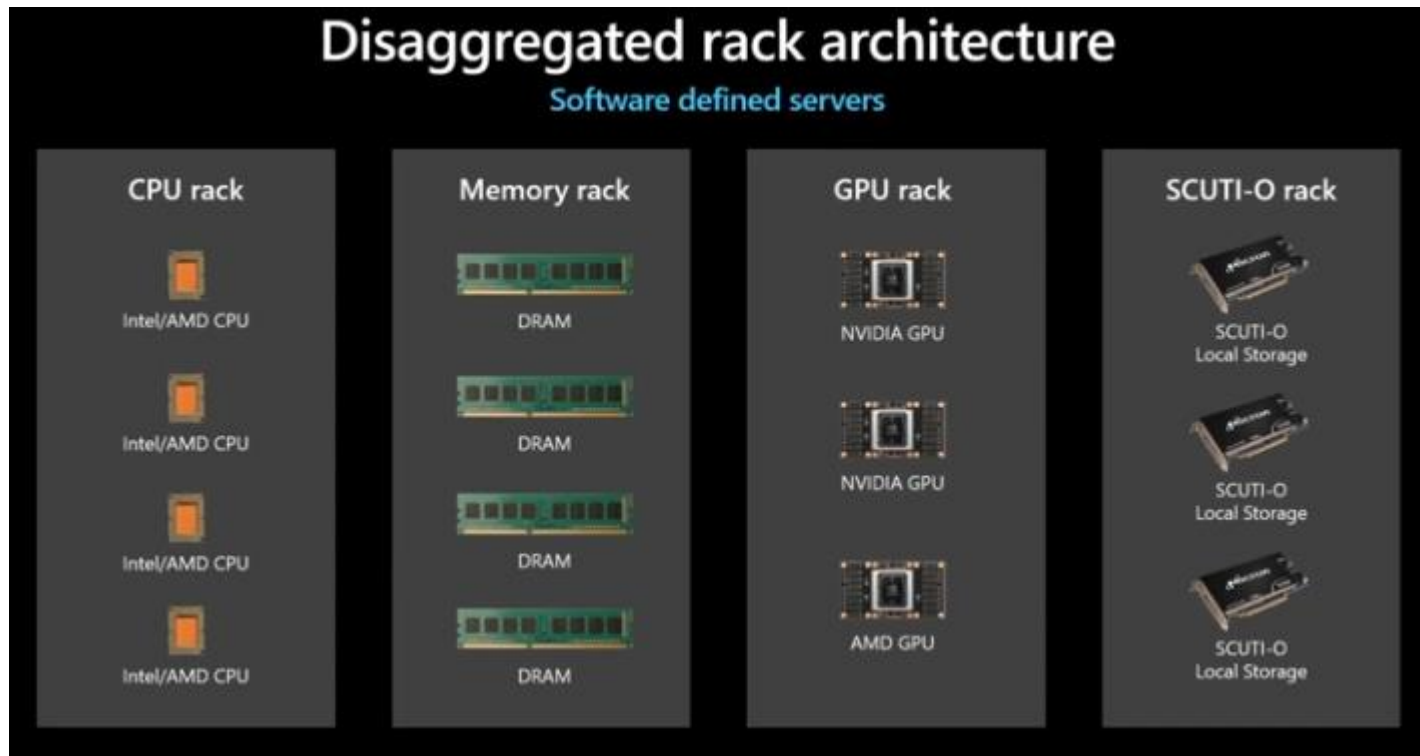
- Virtualization is not enough to improve efficiency
- Need the ability to dynamically allocate CPU resources across servers and racks, allowing admins to quickly migrate resources to address the shifting demand
- Drive 100-300% better utilization for virtualized WLs, and 200-600% for bare-metal WLs.

Software defined DCs

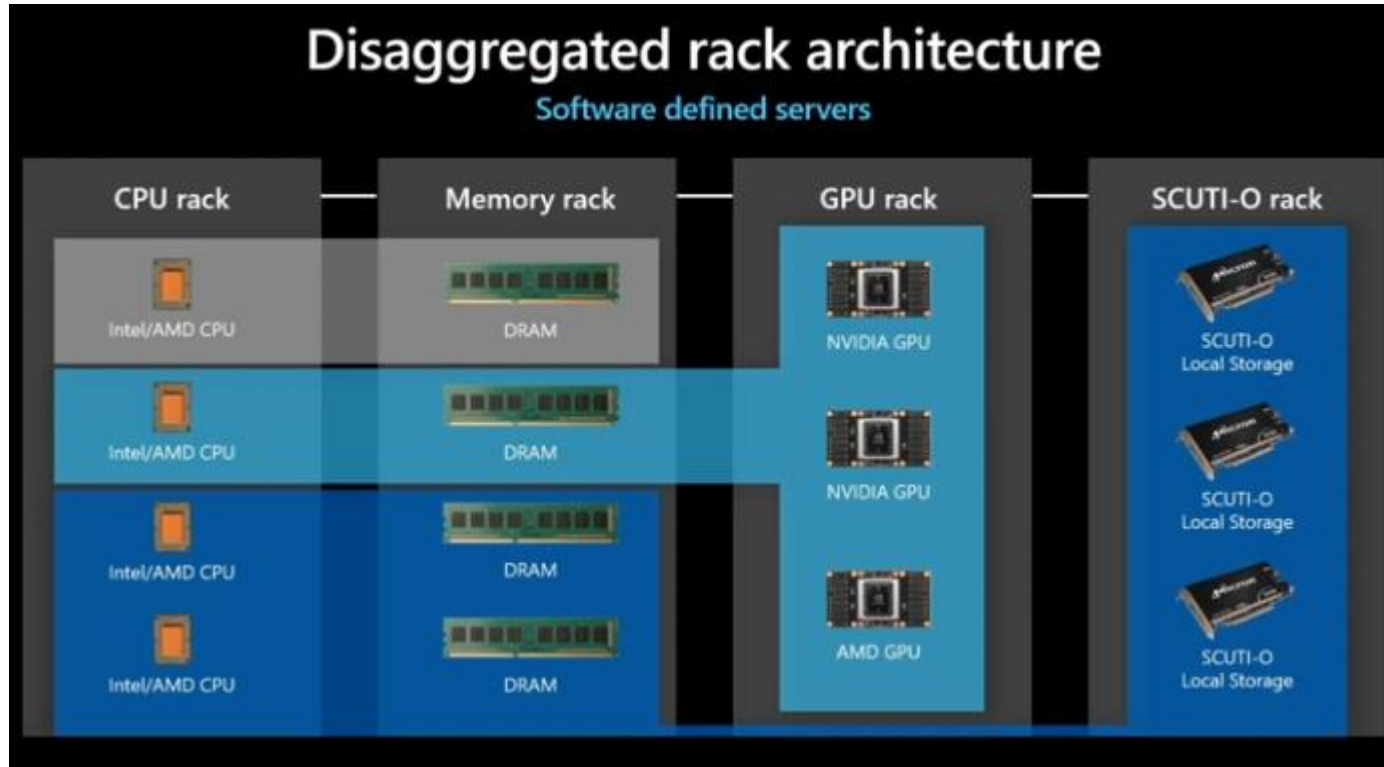
Traditional rack architecture



Disaggregation across racks

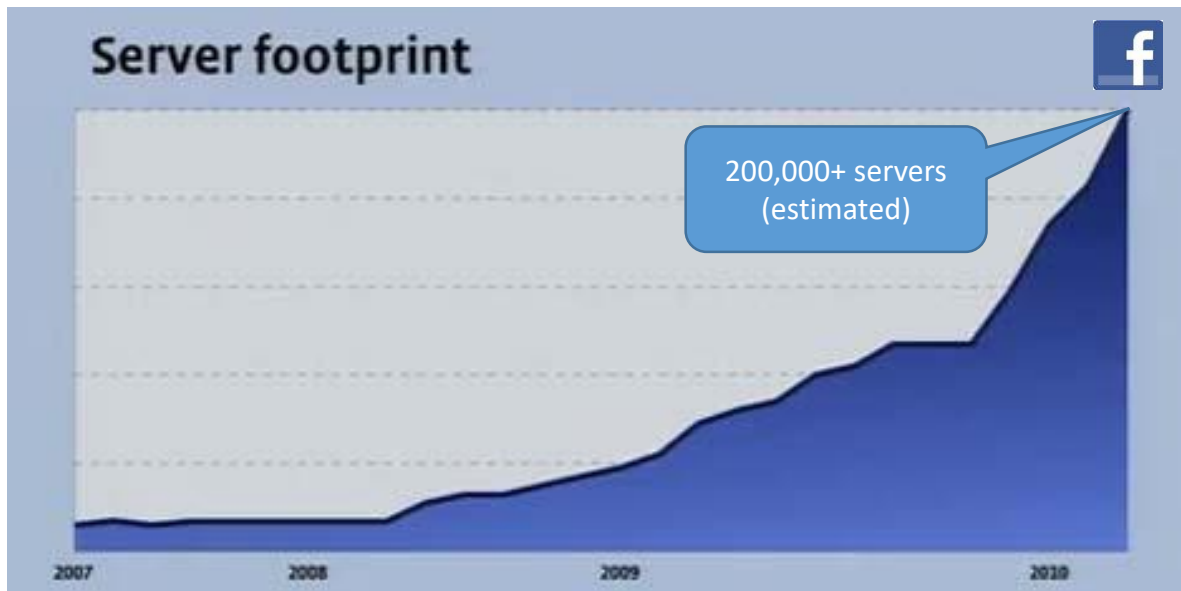


Software defined Servers



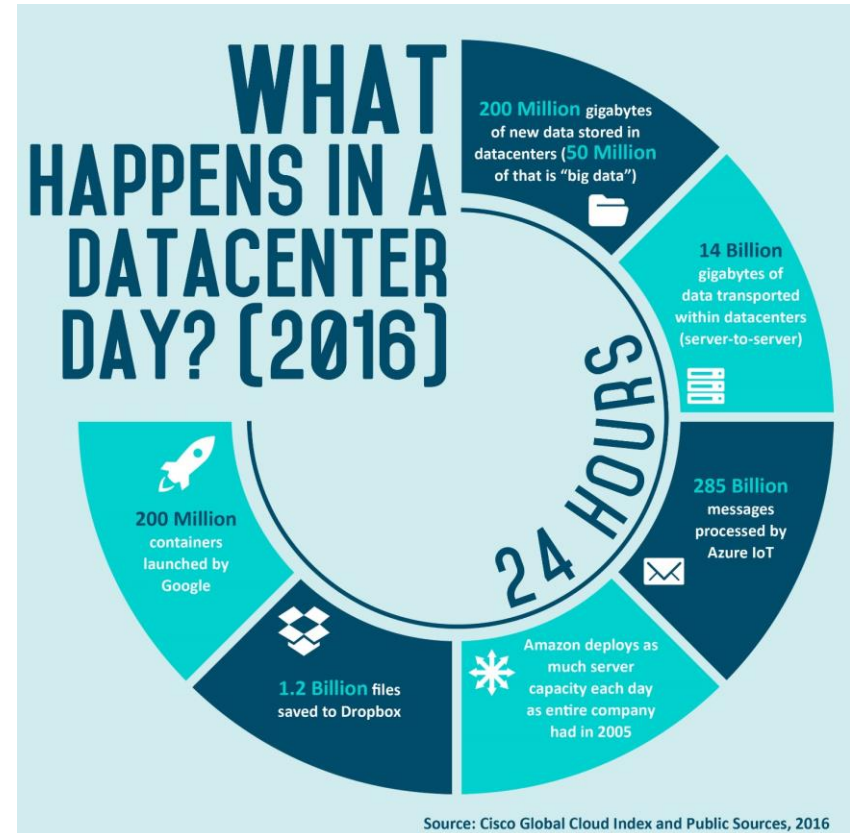
Challenge 5: Managing scale and growth

- In 2016, Gartner estimated that Google has 2.5 million servers
- In 2017, Microsoft Azure was reported to have more than 3 million servers.



Size and growth of DC (2016-2020)

- The scale and complexity of DC operations grows constantly.
- By 2020, Cisco estimated that we would have 600 million GB of new data saved each day
- So the volume of BigData by 2020 was estimated to be as much as all of the stored data in 2016.



Challenge 6: networking at scale



[David Samuel Robbins, gettyimages.ch]



[@AlexCWheeler, Twitter]

Challenge 6: networking at scale (cont.)

- **Building the right abstractions to work for a range of workload at hyper-scale.**

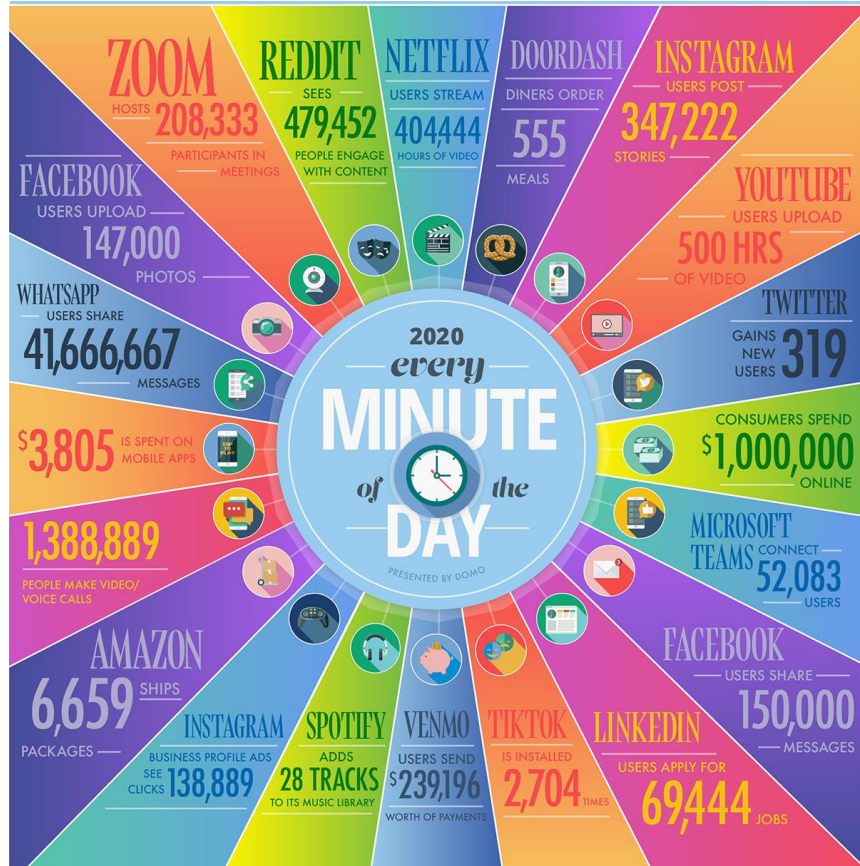


Software Defined Networking (SDN)

- **Within DC, 32 billion GBs will be transported in 2020**
 - src: Cisco's report 2016-2020
- **"Machine to machine" traffic is orders of magnitude larger than what goes out on the Internet**
 - Src: Jupiter Rising: A Decade of Clos Topologies and Centralized control in Google's Datacenter network (ACM SIGCOMM'15)

Cloud Computing Overview

Big Data and the need for Cloud



The world's internet population is growing significantly year over year. As of April 2020, the internet reaches 59% of the world's population and now represents 4.57 billion people — a 6% increase from January 2019.



<https://www.visualcapitalist.com/every-minute-internet-2020/>

Cloud and Cloud computing

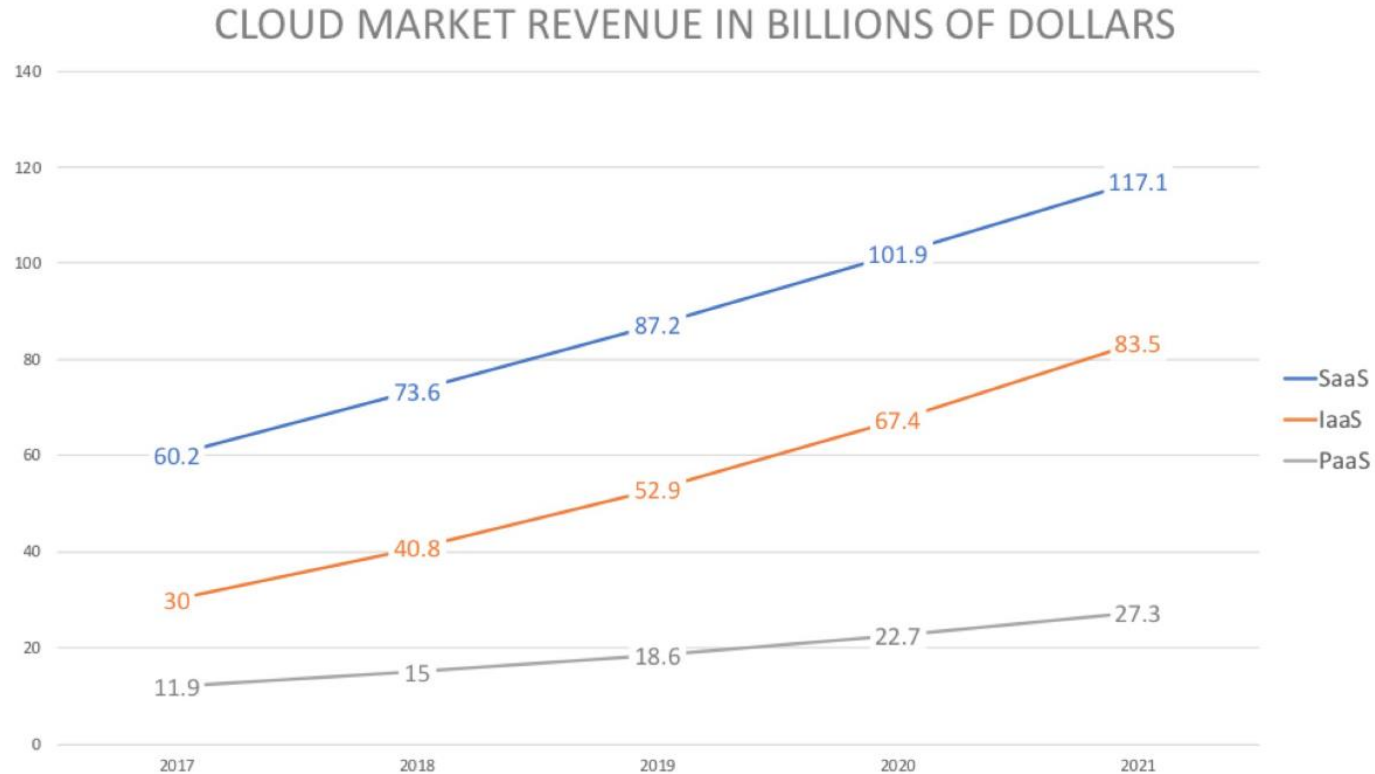
- Datacenter hardware and software that the vendors use to offer the computing resources and services.
- The cloud has a large pool of easily usable **virtualized** computing **resources**, development **platforms**, and various services and **applications**.
- Cloud computing is the delivery of **computing as a service**.
- The **shared resources, software,** and **data** are provided by a provider



Cloud Computing

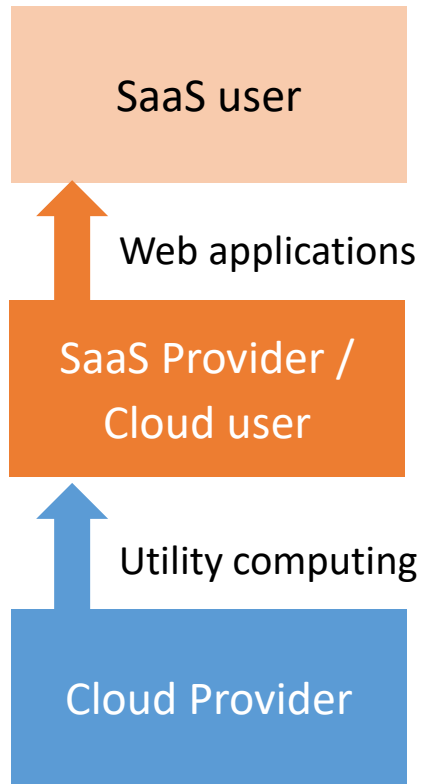
- Datacenters are vendors that rent servers or other computing resources (e.g., storage)
 - Anyone (or company) with a “credit card” can rent
 - Cloud resources owned and operated by a third-party (cloud provider).
- Fine-grained pricing model
 - Rent resources by the hour or by I/O
 - Pay as you go (pay for only what you use)
- Can vary capacity as needed
 - No need to build you own IT infrastructure for peak loads

Cloud market revenue in billions of dollars



source: <https://www.skyhighnetworks.com/cloud-security-blog/microsoft-azure-closes-iaas-adoption-gap-with-amazon-aws/>

Cloud Roles



Application users

Cloud Users

- Software / websites that serve real users
 - Netflix, Pinterest, Instagram, Spotify, Airbnb, Lyft, Slack, Expedia
 - Data analytics, machine learning, and other data services
 - Databricks, Snowflake, GE Healthcare
 - Mobile and IoT backends
 - Snapchat, Zynga (AWS → zCloud → AWS)
 - Datacenter's own software
 - Google Drive/One Drive, search, etc.
-
- **Cloud Providers**
 - Companies with large DCs
 - Amazon AWS, Microsoft Azure, Google Cloud Platform, Alibaba Cloud, ...

Types of Cloud Computing

- **Public vs. Private**

- Public: resources owned and operated by the one organization aka the cloud vendor
- Private: Resources used exclusively by a single business or organization

- **On-premise vs. Hosted:**

- On-premise (on-prem): resources located locally (at a datacenter that the organization operates)
- Hosted: resources hosted and managed by a third-party provider

Private cloud can be both on-prem and hosted (virtual private cloud)

Types of Cloud Computing (cont)

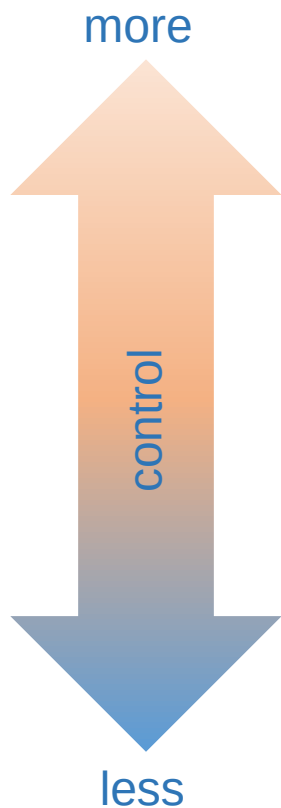
- **Hybrid cloud**

- Combines public and private clouds, allows data and applications to be shared between them.
- Better control over sensitive data and functionalities
- Cost effective, scales well and is more flexible

Multi-Cloud

- Use multiple clouds for an application / service
- Avoids data lock-in
- Avoids single point of failure
- But, need to deal with API differences and handle migration across clouds

Cloud service models (XaaS)



- **Infrastructure as a Service (IaaS)**

- Rent IT infrastructure – servers and virtual machines (VMs), storage, network, firewall, and security

- **Platform as a Service (PaaS)**

- Get on-demand environment for development, testing and management of software applications: servers, storage, network, OS, databases, etc.

- **Serverless, Function as a Service (FaaS)**

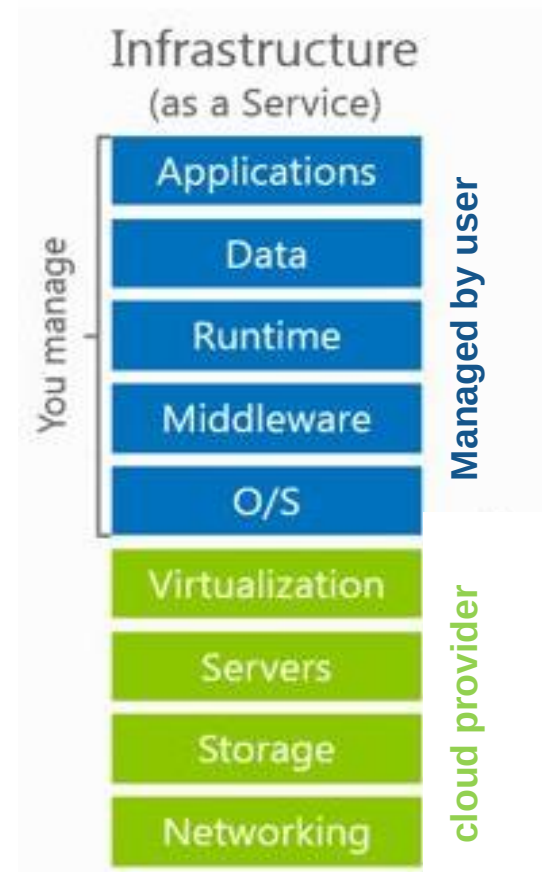
- Overlapping with PaaS, serverless focuses on building app functionality without managing the servers and infrastructure required to do so.
- Cloud vendors provides set-up, capacity planning, and server management.

- **Software as a Service (SaaS)**

- Deliver software applications over the Internet, on demand.
- Cloud vendor handles software application and underlying infrastructure

Infrastructure as a Service

- Immediately **available computing infrastructure**, provisioned and managed by a cloud provider.
- Computing **resources pooled** together to serve multiple users / tenants.
- Computing resources include: storage, processing, memory, network bandwidth, etc.



src image from Microsoft Azure

Platform as a Service

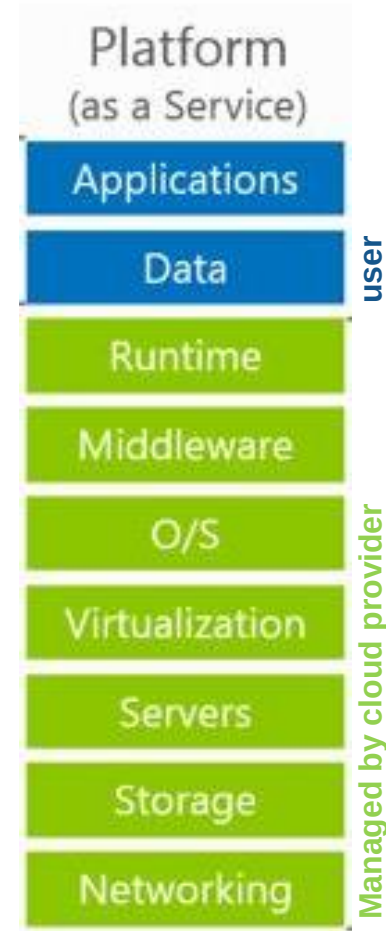
- Complete development and deployment environment.
- Includes system's software (OS, middleware), platforms, DBMSs, BI services, and libraries to assist in development and deployment of cloud-based applications.
- Examples:



ORACLE®
CLOUD PLATFORM



- What is serverless computing then?



src image from Microsoft Azure

Software as a Service



Google™ Apps



Dropbox



Cloud pros and cons

- **User's benefits:**

- Elimination of up-front commitment
- Speed – services are provided on demand
- Global scale and elasticity
- Productivity
- Performance and security
- Customizability
- Ability to pay for use of computing resources on a short-term basis (as needed)

User's concerns:

- Dependability on network and internet connectivity
- Security and privacy
- Cost of migration
- Cost and risk of vendor lock-in